Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)		200 [g]		1		1	1	1
		(ii)		$\frac{9.3 - 8.9}{2}$ (1) = 0.2 [cm] (1) **ecf** for incorrect mass selected in (i) except for 500 [g]		2		2	2	2
	(b)			<u>Reduce</u> the distance between the ruler and the spring / use a pointer			1	1		1
	(c)			Calculate mean <u>extension</u> for each {force / mass} (1) Plot a graph of force against [mean] extension (1) Determine the gradient – this is the spring constant (1) **Alternative:** Calculate mean <u>extension</u> for each {force / mass} (1) Determine k for each force from $F = kx$ (1) Add up the <u>5 values</u> for k and use them to calculate the mean (1) Don't accept reference to 6 values **Alternative:** Determine k from $F = kx$ (1) Do this for all extensions (1) Add up the <u>10 values</u> for k and use them to calculate the mean (1)		3		3		3
	(d)			Data are accurate (1) because the spring constant is close to the true value (1) Accept the converse argument i.e. Data are <u>slightly</u> inaccurate (1) because the spring constant is not exactly the true value (1)			2	2		2
				Question 5 total		**6**	**3**	**9**	**3**	**9**